

Week 3 Technical Report

1. Introduction

In Week 3 of my internship, I worked on TensorFlow/Keras models, text classification, prompt engineering, and AI-assisted coding. I implemented and tested different models using Google Colab and documented their performance.

2. TensorFlow/Keras Model Comparison

I used the MNIST handwritten digit dataset to build two neural network models.

Model 1

- Simple neural network
- One hidden Dense layer
- Test Accuracy: **97.36%**
- Test Loss: **0.0847**

Model 2

- Improved neural network
- Two Dense layers with Dropout
- Test Accuracy: **97.42%**
- Test Loss: **0.0894**

Model 2 achieved slightly higher test accuracy than Model 1.

3. Text Classification Project

I developed a movie review sentiment classification model using the IMDB dataset.

The model classifies reviews into two categories:

- Positive
- Negative

The model was developed using TensorFlow and Keras with an Embedding layer, Global Average Pooling, and Dense layers.

Result

- Test Accuracy: **87.19%**
- Test Loss: **0.3246**

The model was also tested with new movie reviews to verify its predictions.

4. Prompt Engineering

I created three versions of a prompt for explaining CNN:

1. Basic prompt
2. Improved prompt
3. Structured prompt

The prompts were compared based on clarity, relevance, usefulness, and overall quality.

The structured prompt produced the most specific and useful response because it clearly defined the role, audience, requirements, and expected format.

5. AI-Assisted Coding

I used AI assistance while developing a small **Movie Review Sentiment Analyzer** application.

The application was created using Gradio and connected to the trained sentiment classification model.

The app accepts a movie review and displays:

- Sentiment
- Confidence score

The application was tested using both positive and negative reviews.

6. Conclusion

Week 3 helped me understand practical applications of TensorFlow/Keras and text classification. I also learned how prompt structure affects AI-generated results and how AI can assist in software development.

Overall, this week improved my understanding of Deep Learning, NLP, prompt engineering, and AI-assisted coding.